Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is a ray infinite in one direction? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Does a line extend infinitely in both directions? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

How many points are needed to define a unique line? Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A reflection maps segment AB to A′B′. Is AB congruent to A′B′? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A 40° angle is rotated about a point. What is the measure of the image angle? Student's answer: 80°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Does a dilation preserve distances? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$\triangle ABC \cong \triangle DEF$, and $\angle A = 50°$. What is $\angle D$? Student's answer: 130°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$\triangle PQR \cong \triangle XYZ$, and $PQ = 7$. What is XY? Student's answer: 14

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In $\triangle ABC \cong \triangle DEF$ with $A \leftrightarrow D$, $B \leftrightarrow E$, and $C \leftrightarrow F$, which side corresponds to AC? Student's answer: DE

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two triangles have sides 3, 4, and 5 in both. Are they congruent? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two triangles have angles 50° and 60° with a side of 4 between those angles in both. Which criterion proves congruence? Student's answer: AAA

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is SSA a valid criterion for proving triangles congruent? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two angles are vertical angles. One measures 70°. What is the other? Student's answer: 110°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two parallel lines are cut by a transversal. An alternate interior angle is 65°. What is the other?
Student's answer: 115°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A point is on the perpendicular bisector of a segment. Is it equidistant from the endpoints?
Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 10 · G.CO.10

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A triangle has angles of 40° and 75°. What is the third angle? Student's answer: 75°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A triangle has a base of 12. What is the length of its midsegment parallel to the base? Student's answer: 12

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An exterior angle of a triangle is 115°. One remote interior angle is 50°. What is the other remote interior angle? Student's answer: 165°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a parallelogram, one angle is 110°. What is the opposite angle? Student's answer: 70°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

The diagonals of a parallelogram meet at O. If AC = 14, what is AO? Student's answer: 14

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a parallelogram, one angle is 110°. What is an adjacent angle? Student's answer: 110°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A dilation with scale factor 3 maps a side of length 4 to a new side. What is the new length?
Student's answer: 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Are two triangles similar if two pairs of angles are congruent? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a similar pair, a side of length 3 in the small triangle matches a side of 6 in the large triangle. A side of 4 in the small triangle is what in the large? Student's answer: 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Similar triangles have AB = 4, DE = 6, and BC = 8. What is EF if EF corresponds to BC? Student's answer: 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line parallel to BC crosses AB at D and AC at E. AD = 3, DB = 6, and AE = 4. What is EC? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a right triangle, the altitude to the hypotenuse splits it into segments of 4 and 9. What is the altitude? Student's answer: 13

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Triangles have two sides and the included angle congruent. Which method proves congruence?
Student's answer: SSA

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Similar triangles have sides 5 and 10 matching, and a side of 14 in the second triangle matches a side of x in the first. Find x. Student's answer: 9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In similar triangles, AB = 3 and DE = 6, and AC = 5. What is DF? Student's answer: 5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a right triangle, the opposite side to angle A is 3 and the hypotenuse is 5. What is sin A?
Student's answer: 4/5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 10 · G.SRT.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a right triangle, the adjacent side to angle B is 8 and the hypotenuse is 17. What is cos B?
Student's answer: 15/17

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is tan 45°? Student's answer: $\sqrt{2}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A right triangle has legs 3 and 4. What is the hypotenuse? Student's answer: 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A right triangle has opposite side 5 and hypotenuse 10 for angle A. What is angle A? Student's answer: 60°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A 10-foot ladder makes a 60° angle with the ground. How high does it reach? Student's answer: 5 feet

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A central angle measures 80°. What is the inscribed angle that intercepts the same arc?
Student's answer: 80°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An inscribed angle measures 35°. What is the intercepted arc? Student's answer: 35°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An inscribed angle is drawn with its sides on a diameter. What is its measure? Student's answer: 180°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the arc length of a 90° arc in a circle with radius 6. Student's answer: 6π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the area of a 60° sector in a circle with radius 3. Student's answer: 9π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the arc length of a 120° arc in a circle with radius 9. Student's answer: 12π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rectangle has vertices A(0, 0), B(4, 0), C(4, 3), and D(0, 3). What is the length of diagonal AC?
Student's answer: 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the slope of the line through A(1, 2) and B(5, 10)? Student's answer: 1/2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Are the lines $y = 2x + 1$ and $y = 2x - 3$ parallel? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the midpoint of segment AB with A(0, 0) and B(10, 0). Student's answer: (4, 0)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the point 1/3 of the way from A(0, 0) to B(12, 0). Student's answer: (3, 0)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the point that divides A(2, 0) to B(8, 0) in a 2:1 ratio, measured from A. Student's answer: (5, 0)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rectangle has vertices (0, 0), (5, 0), (5, 2), and (0, 2). What is its perimeter? Student's answer: 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A right triangle has vertices (0, 0), (6, 0), and (0, 8). What is its area? Student's answer: 14

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A square has vertices (1, 1), (4, 1), (4, 4), and (1, 4). What is its area? Student's answer: 12

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the circumference of a circle with radius 4. Student's answer: 4π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the volume of a cylinder with radius 3 and height 5. Student's answer: 15π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the volume of a cone with radius 3 and height 6. Student's answer: 54π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A box is 4 by 5 by 6 units. What is its volume? Student's answer: 15

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A cylinder has volume 200π and radius 5. What is its height? Student's answer: 40

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A sphere has volume 36π. What is its radius? Student's answer: 9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which solid best models a soup can? Student's answer: cone

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which solid best models a basketball? Student's answer: cylinder

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rectangular garden is 12 feet by 7 feet and needs fencing around its edge. How much fencing is needed? Student's answer: 19 feet

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.